Charges formed in ions

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt									

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Key:

Colour	Charge
	1+
	2+
	3+
	4+
	1-
	2-
	3-

Create a colour key grouping the ions by their different charges.

Then find the ions on the Periodic Table and colour them in.

	Special
⧸	2 or more +ve charges (II, III, IV)

Go through the first18 elements and write in how many electrons are needed to fill the last shell and thus make the atom stable. The first 2 are examples.

Element	Electrons needed to stabilise atom	Element	Electrons needed to stabilise atom	Element	Electrons needed to stabilise atom
Hydrogen	1	Nitrogen		Aluminium	
Helium	0	Oxygen		Silicon	
Lithium		Fluorine		Phosphorous	
Beryllium		Neon		Sulphur	
Boron		Sodium		Chlorine	
Carbon		Magnesium		Argon	

Looking at the table above, let's order the first 18 elements into the table below.

Elements which need **1** more electron to be stable.	Elements which need **2** more electrons to be stable.	Elements which need **3** more electrons to be stable.	**None needed** Stable elements